

Overlapping Call Density Regression for Acoustic Population Size Estimation

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Abstract—Passive acoustic monitoring enables population size estimation from audio recordings, but existing methods show higher estimation error in large populations, because larger groups tend to produce more frequent and severe call overlap that distorts the acoustic evidence such methods rely on. To address this problem, we propose a population estimation system based on overlapping call density regression. The proposed system employs a convolutional recurrent regressor that predicts a call density for each audio segment from its log-mel spectrogram, trained with a square-root-domain mean squared error loss to reduce the dominance of high-density samples during training. The segment-level density predictions are then aggregated into the mean and standard deviation, which are mapped to the final population estimate through linear regression. We conduct experiments on the real-world dataset from the BioDCASE 2026 Bird Counting Challenge, using the Greater flamingo as a representative flock-calling species. Experimental results show that the proposed system substantially outperforms the challenge baseline. Under leave-one-out cross-validation on the four development-set aviaries, the mean absolute error (MAE) is reduced from 23.0 to 1.5. In the official challenge evaluation, the proposed system ranked first among all submissions.

Index Terms—Bioacoustics, bird counting, population size estimation, Greater flamingo, flock-calling species

1. INTRODUCTION

Bird population size estimation [1]–[5] is a fundamental task in ecological monitoring and conservation, providing essential information for assessing habitat quality and population trends. The growing availability of passive acoustic monitoring (PAM) [6]–[8] has made large-scale, long-term acoustic recordings increasingly accessible for this task [9]–[12]. However, estimating the population size of a target species from such recordings, which typically contain background noise and calls from non-target species, remains a challenging problem.

Existing methods for estimating population size from PAM recordings fall into two categories, detection-based and detection-free approaches. Detection-based approaches [13]–[16] first detect a countable acoustic unit within short analysis windows, such as an individual vocal signature or a call event, and then map a summary statistic of these detections to population size. This pipeline has been used to estimate bat population density from call-count statistics [14], to characterize Savannah Sparrows population through the variability of vocal activity [15], and to predict population size across several bird species with a confidence-weighted detection rate derived from detected call events [16]. These methods assume that acoustic events in PAM recordings are distinguishable. However, acoustic overlap has been found to substantially undermine this assumption, leading to systematic underestimation of population size. Detection-free approaches [17]–[19] bypass the detection stage and instead predict population size or abundance-related proxies directly from acoustic features. For example, gradient boosting regressors trained on generated chorus mixtures have been used to predict Greater

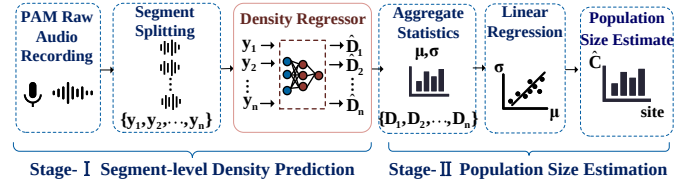


Fig. 1: Overview of the proposed two-stage population estimation system.

flamingo population size directly. When applied to real recordings from aviaries with much larger populations, predictions saturated near the upper bound of the training range, distinguishing relative differences in vocal density but not recovering absolute population size [17]. Overall, existing methods perform well when population size is small or when vocalizations remain distinguishable, but they share a common bottleneck under severe acoustic overlap, which typically arises as population size grows.

To address this problem, we propose a two-stage population estimation system, as illustrated in Fig. 1. In the first stage, each PAM segment is processed independently by a density regressor, which predicts the density of target-species calls rather than their count, since its output is a continuous scalar. In the second stage, the segment-level density estimates are aggregated into the site-level mean and standard deviation, which are mapped to the final population estimate through a linear model calibrated on labeled sites. The proposed system follows the detection-free paradigm and differs from prior approaches [17] in three respects. First, the density regressor adapts a CRNN backbone [20] by replacing the original frame-level multi-label classification output with a temporal pooling operation and a regression head, producing a segment-level scalar prediction. Second, the regressor is trained with a square-root-domain mean squared error (MSE) loss designed to stabilize gradients under high-overlap samples. Third, prior work trained the regressor on generated mixtures containing up to 10 individuals, whereas we investigate a wider range of overlapping calls, up to a maximum of 200.

We evaluate the proposed system on real-world aviary PAM recordings from the BioDCASE Bird Counting challenge [21], targeting the Greater flamingo, a flock-calling species that frequently vocalizes synchronously in large groups. Evaluation results show that the proposed system reduces the official baseline’s mean absolute error (MAE) from 69.0 to 20.0, ranking first overall among all submitted systems. The main contributions of this work are threefold: (1) We propose a population estimation method based on segment-level density prediction, which recovers absolute population size from PAM recordings with severe call overlap. (2) We introduce a square-root-domain MSE loss for training the density regressor, and show through an ablation study that it achieves lower estimation error than standard loss functions such as MSE and MAE. (3) We find that the range of call counts used to generate training data has a non-

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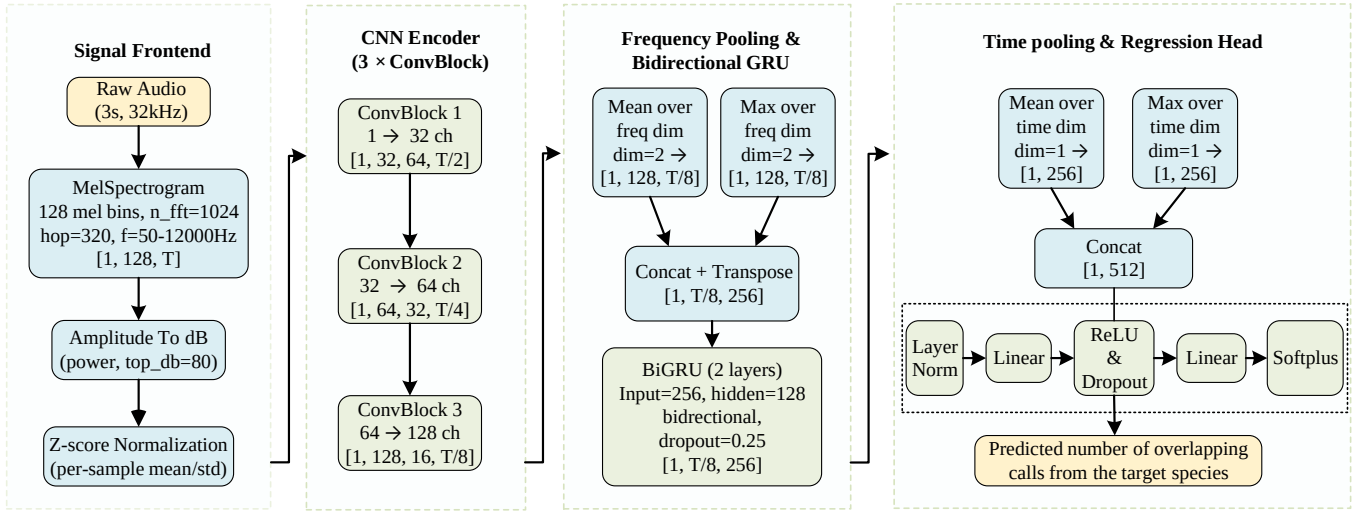


Fig. 2: Architecture of the density regressor for segment-level call density prediction.

monotonic effect on estimation accuracy. A range that is too narrow prevents the model from learning to estimate high-density segments accurately, while a range that is too wide increases the difficulty of the regression task itself and reduces overall estimation accuracy.

2. PROPOSED METHOD

We formulate population size estimation from acoustic recordings as a two-stage regression problem. Consider an audio recording that contains calls from a target species with an unknown population size C , together with calls from other species and background noise. The recording is divided into N segments $\{y_1, \dots, y_N\}$ of fixed duration. Each segment y_n has a vocal density D_n , defined as the number of overlapping calls from the target species it contains. In the first stage, we estimate a segment-level vocal density $\hat{D}_n$ from each segment y_n . In the second stage, we estimate the population size $\hat{C}$ from the segment-level estimates $\{\hat{D}_n\}_{n=1}^N$.

2.1. Stage-I Segment-Level Density Regression

The density regressor is built on a CRNN backbone. The original CRNN architecture [20] was designed for frame-level polyphonic sound event detection, producing a sequence of per-frame, per-class activity probabilities. We adapt this architecture to segment-level regression by replacing the frame-wise output with a temporal pooling operation followed by a regression head, so that the network produces one scalar density estimate for the entire segment instead of a sequence of frame-level predictions.

As illustrated in Fig. 2, the waveform of each segment y_n is transformed into a log-mel spectrogram using a mel filterbank followed by logarithmic compression. The resulting feature is then normalized using per-sample z-score normalization to reduce scale variation across segments and improve training stability. A convolutional encoder then processes the spectrogram, progressively reducing both the frequency and time dimensions. The remaining frequency bins are further pooled through mean and max pooling and concatenated, and the resulting sequence is passed to a bidirectional recurrent layer [22], [23] to model temporal context across the segment. The recurrent outputs are aggregated over time through mean and max pooling, yielding a segment-level representation. This representation is mapped to the density estimate $\hat{D}_n$ by a regression head consisting of two

linear layers separated by a ReLU activation and dropout, with layer normalization applied to the input and a Softplus activation applied to the final output.

The density regressor is trained by minimizing a mean squared error computed in the square-root domain, referred to as Sqrt-MSE, which reduces the dominance of high-count samples during optimization. Let D_n and $\hat{D}_n$ denote the ground-truth and predicted call densities of the n -th audio segment, respectively. The loss is defined as

$$L_{\sqrt{\text{MSE}}} = \left(\sqrt{\hat{D}_n} - \sqrt{D_n} \right)^2. \quad (1)$$

2.2. Stage-II Population Size Estimation

In the second stage, the segment-level density estimates $\{\hat{D}_n\}_{n=1}^N$ produced by Stage-I are aggregated into a compact site-level descriptor, which is then mapped to the population size estimate $\hat{C}$. Specifically, we summarize the segment-level estimates using their mean μ and standard deviation σ , which characterize the average calling activity and its variability at the site, respectively. These two statistics are mapped to the population size through a linear regression model:

$$\hat{C} = w_1 \mu + w_2 \sigma + b, \quad (2)$$

where w_1 , w_2 , and b are learnable parameters fitted by ordinary least squares on sites with known population sizes. Given limited site-level supervision, this low-complexity mapping helps prevent overfitting. The resulting values are $w_1 = 88.86$, $w_2 = -34.67$, and $b = 38.46$.

3. EXPERIMENT SETTINGS

3.1. Training Dataset

Real recordings annotated with exact call counts are limited, and existing annotations typically exclude regions with heavily overlapping calls because these regions are difficult to label accurately. We therefore generated a dataset to train the segment-level density regressor described in Section 2.1. The dataset consisted of Greater flamingo call snippets and calls from other bird species collected from Xeno-Canto¹, together with background audio extracted from the challenge development-set aviary recordings [21].

¹<https://xeno-canto.org>

Table 1: Official leaderboard on the BioDCASE 2026 Bird Counting Challenge evaluation set. Only the top three submissions and the official baseline are shown. The proposed system corresponds to submission Yang_WHU_task6_1.

Submission	Rank	Score↓	MAE↓			RMSE↓	R^2 ↑
			Flamingo	Quelea	Ibis		
Yang_WHU_task6_1	1	9.2	20.0	4.5	3.0	13.0	0.97
Leon_UFAM_task6_3	2	13.3	23.5	6.0	10.5	18.0	0.94
You_PKU_task6_2	3	22.7	61.5	3.5	3.0	48.7	0.59
official_aria_baseline	4	24.5	69.0	1.5	3.0	51.4	0.55

Table 2: Leave-one-out cross-validation results on the development set. Positive and negative signs indicate overestimation and underestimation, respectively.

	Aviary-2 (Dev)	Aviary-4 (Dev)	Aviary-5 (Dev)	Aviary-6 (Dev)	MAE↓	RMSE↓	R^2 ↑
Ground truth	107	161	52	52	–	–	–
Baseline	+9	−64	+5	+14	23.0	33.2	0.46
Ours	+1	−2	+2	−1	1.5	1.6	0.99

Greater Flamingo Call Snippets: The extraction procedure for target species calls refers to [17]. We collected 429 wild-recording files of Greater flamingo vocalizations from Xeno-Canto. Each recording was split into 3-second segments with 50% overlap. Two independent detectors, BirdNET [24] and PERCH v2 [25], were applied to every segment. Only segments with detection confidence scores of at least 0.9 from both detectors were retained, and adjacent overlapping segments among them were merged into continuous audio spans. Call events were then extracted from these merged recordings using an RMS energy envelope with adaptive boundary detection. The RMS envelope was computed using a frame length of 20 ms, a hop size of 10 ms, and 7-frame smoothing. This procedure produced 2,587 Greater flamingo call snippets in total, with durations between 0.1 s and 1.5 s.

Other-bird call snippets: Calls from 18 bird species other than the Greater flamingo, selected based on the species reported in [21], were collected from Xeno-Canto. Using the same extraction procedure as for Greater flamingo, we obtained 337 call snippets, which were used as interference calls in subsequent data generation.

Background audio: We used the detection results of the BirdNET and ARIA species detectors provided by the challenge to filter candidate background segments. Specifically, a 3-second segment was retained as a candidate only if neither detector indicated a Greater flamingo call. This filtering yielded 47,531 candidate background segments. As the filtering criterion excludes only the Greater flamingo, these segments may still contain calls from other bird species.

Mixing protocol: All source materials were resampled to 32 kHz single-channel audio before mixing. Each 3-second training segment was generated by mixing Greater flamingo call snippets, interference snippets from other bird species, and background audio.

$$x = \sum_{i=1}^K s_i^{\text{flamingo}} + \sum_{j=1}^M s_j^{\text{interference}} + s_{\text{bg}}, \quad (3)$$

where s_i^{flamingo} is the i -th Greater flamingo call snippet, $s_j^{\text{interference}}$ is the j -th interference snippet from another bird species, and s_{bg} is the background segment. K is the number of Greater flamingo calls in the segment and thus the segment-level label, whereas M is randomly sampled from $\{0, 1, 2, 3\}$. Each snippet was assigned a random onset within the segment. This random placement allowed call events to overlap.

To simulate variability in real recordings, each audio component was assigned a random gain in the range of $[-10, +10]$ dB during mixing. Unless otherwise specified, we adopted a target-species call count

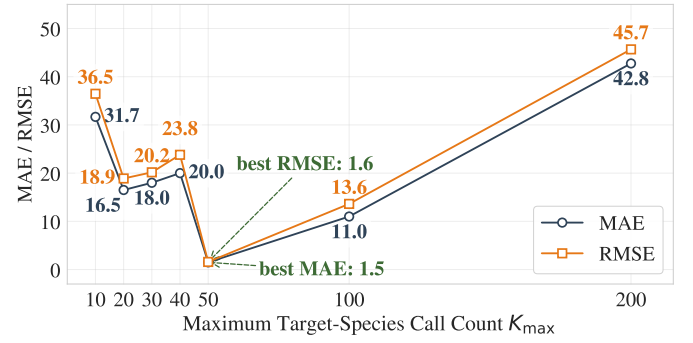


Fig. 3: Development-set MAE and RMSE for different target-species call count ranges $[0, K_{\max}]$. Each $K_{\max}$ is used to generate a separate training set for the density regressor.

range of $[0, 50]$. For each count value in this range, we generated 150 training segments and 10 validation segments, resulting in a balanced dataset with diverse call-density and interference conditions. To prevent data leakage, source materials were split into training and validation subsets at the recording level, and segments were generated only within the corresponding subset.

3.2. Evaluation Dataset

We evaluated the proposed system using the BioDCASE Bird Counting Challenge dataset [21]. Its development set comprises 140,899 3-second audio segments collected from six aviaries in European zoos using passive acoustic monitoring devices, while its evaluation set contains approximately 380,000 3-second audio segments from 10 held-out aviaries. This study focuses on the Greater flamingo. In the development set, this species is present in four aviaries, and the recordings also include other non-target bird species and environmental noise. In the evaluation set, two aviaries contain Greater flamingos.

3.3. Evaluation Metrics

We employed three metrics to assess the performance of the proposed system. MAE [26] is the average absolute difference between the predicted and true population sizes. RMSE is the square root of the average squared difference between the predicted and true population sizes, and therefore penalizes large errors more heavily than MAE. The coefficient of determination (R^2) [27] is the proportion of variance in the true population sizes explained by the predictions.

3.4. Training Configuration

The segment-level density regressor is trained for up to 100 epochs with a batch size of 32. We use the AdamW [28] optimizer with an initial learning rate of 3×10^{-4} and a weight decay of 1×10^{-4} . The learning rate is reduced by a factor of 0.5 when the validation MSE does not improve for 3 epochs, with a minimum learning rate of 1×10^{-6} . Early stopping is applied if the validation MSE does not improve for 10 epochs. Gradient clipping is used with a maximum norm of 5.0. During training, SpecAugment [29] is applied to the input features using 2 frequency masks with a maximum width of 12 mel bands and 2 time masks with a maximum width of 20 frames. All experiments are conducted on a single NVIDIA RTX 4090 GPU. For aviary-level population estimation, we fit the linear regression model using a leave-one-out [30] protocol over the four Greater flamingo aviaries in the development set.

Table 3: Ablation study of loss functions on the development set.

Loss Function	Formulation	Aviary-2 (Dev)	Aviary-4 (Dev)	Aviary-5 (Dev)	Aviary-6 (Dev)	MAE↓	RMSE↓
MAE	$ D - \hat{D} $	+3	−6	+12	−5	6.5	7.3
MSE	$(D - \hat{D})^2$	−19	+57	+24	−	33.3	37.4
Sqrt-MAE	$ \sqrt{D} - \sqrt{\hat{D}} $	−5	+11	−4	+3	5.8	6.5
Sqrt-MSE	$(\sqrt{D} - \sqrt{\hat{D}})^2$	+1	−2	+2	−1	1.5	1.6

4. RESULTS AND DISCUSSIONS

4.1. Overall Performance

Table 1 presents the official leaderboard results on the evaluation set. Score denotes the ranking metric, the MAE pooled across all (aviaries, species) pairs. Our proposed system ranked first overall and achieved the lowest MAE for the Greater flamingo among all submitted systems, reducing it from 69.0 to 20.0. The Quelea and Ibis results are included only for completeness, as ranking score, RMSE, and R^2 are computed jointly over all three target species. For these two species, our submission used the official baseline system with hyperparameters tuned on the development set. Across all submissions, the MAE for the Greater flamingo is substantially higher than for Quelea and Ibis, consistent with its synchronous flock-calling behavior that produces more call overlap.

Table 2 reports the leave-one-out cross-validation results on the four Greater flamingo aviaries in the development set. The proposed system reduces the overall MAE from 23.0 for the baseline to 1.5, and improves R^2 from 0.46 to 0.99. The largest improvement occurs on Aviary-4 (161 individuals), where the baseline underestimates the population by 64 individuals while our system’s error is only 2. Across the four aviaries, population sizes range from 52 to 161 individuals. The error of the proposed system remains within 2 individuals across all of them, indicating that it generalizes well across different population sizes.

It is worth noting that the Greater flamingo population sizes in Table 1 were estimated using a Stage-II model fitted on all four Greater flamingo aviaries in the development set. The development-set population sizes in Table 2 were instead estimated under the leave-one-out protocol described in Section 3.4, so that each aviary’s population estimate comes from a model calibrated without that aviary’s own ground truth.

4.2. Effect of the Target-Species Call Count Range

We evaluated seven target-species call count ranges, $[0, K_{\max}]$ with $K_{\max} \in \{10, 20, 30, 40, 50, 100, 200\}$, each used to train a separate density regressor with an MSE loss in the square-root domain. The training set size grows as $K_{\max}$ increases, since 150 training segments are generated for each count value in the range. Fig. 3 shows the development-set performance under these seven settings. Both MAE and RMSE follow a U-shaped trend as $K_{\max}$ increases, decreasing sharply at first, reaching their minimum at $K_{\max} = 50$ (MAE=1.5, RMSE=1.6), and then increasing substantially as $K_{\max}$ is further increased. Segments generated with a higher $K_{\max}$ contain more severely overlapping calls. This makes them harder for the regressor to learn from, offsetting the benefit of a larger training set. The resulting degradation reflects the overlapping-call problem motivating this work.

4.3. Effect of the Training Loss

All regressors in this ablation were trained with a fixed target-species call count range of $[0, 50]$. Table 3 compares four loss functions for training the density regressor, MAE, MSE, Sqrt-MAE, and Sqrt-MSE, where the sqrt-variant computes the loss in the square-root domain.

Table 4: Predicted population size under different sampling strategies and sampling ratios on two evaluation aviaries, reported as mean $\pm$ std.

Strategy	Sampling ratio (%)	Aviary-4 (Eval)	Aviary-8 (Eval)
Contiguous	20	41.1 ± 1.7	180.5 ± 20.0
Contiguous	50	40.8 ± 0.9	173.0 ± 5.8
Contiguous	80	40.4 ± 0.5	178.6 ± 1.8
Random	20	40.2 ± 0.7	183.9 ± 1.4
Random	50	40.2 ± 0.4	183.9 ± 0.7
Random	80	40.1 ± 0.2	184.0 ± 0.4

Sqrt-MSE achieves the lowest MAE and RMSE among the four, 1.5 and 1.6 respectively, while MSE yields the largest errors, 33.3 and 37.4. Across the four development aviaries, the system trained with the Sqrt-MSE loss keeps the estimation error within 2 individuals. These results confirm the effectiveness of the proposed Sqrt-MSE loss for population size estimation, and show that this benefit holds across populations of different scales.

4.4. Robustness Analysis

To assess the sensitivity of the proposed system to the amount and temporal continuity of available data, we performed subsampling experiments on two evaluation aviaries. For each aviary, a subset of its 3-second segments was sampled without replacement, at a sampling ratio of 20%, 50%, or 80%. Two sampling strategies were used. Random sampling selected segments uniformly at random from the full set. Contiguous sampling selected a random starting point and drew a contiguous block of segments from that point until the target ratio was reached. Each configuration was repeated 1,000 times.

Table 4 reports the resulting population estimates as mean $\pm$ standard deviation. Across both aviaries, the mean population estimate is stable across all sampling strategies and ratios, while the standard deviation decreases as the sampling ratio increases. At each ratio, random sampling yields a smaller standard deviation than contiguous sampling. This suggests that random sampling is preferable when only a fraction of the available recordings can be used for prediction due to computational or storage constraints. This advantage is most evident for Aviary-8 at a 20% sampling ratio, where the standard deviation is 1.4 for random sampling versus 20.0 for contiguous sampling. These results confirm that the proposed system does not depend on temporal continuity and remains robust even when the available data is substantially reduced.

5. CONCLUSION

This paper proposed a two-stage system for estimating population size from acoustic recordings under severe call overlap. The proposed system predicts segment-level call density from log-mel spectrograms using a convolutional recurrent regressor, and maps aggregated density statistics to the final population estimate through linear regression. The system reduced the development-set MAE for the Greater flamingo from 23.0 to 1.5 under leave-one-out cross-validation, and ranked first in the official BioDCASE 2026 evaluation. Ablation results showed that training with a square-root-domain MSE loss yields a performance gain over standard MSE and MAE, and that the target-species call count range used for training involves a trade-off between the coverage of high-density segments and the difficulty of the regression task. A robustness analysis further showed that the proposed system maintained accurate and stable population estimates under limited and discontinuous data. Future work will explore richer descriptors of the segment-level density distribution beyond its mean and standard deviation, and calibration methods that require fewer or zero labeled sites for the population size mapping.

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